

Netflix Analysis Report: Duration vs Release Year

Introduction

Netflix has grown from a streaming library into a global entertainment platform with thousands of titles spanning many years, genres, and formats. This report matters because content length is one way to see how Netflix production and acquisition patterns have changed over time, which helps explain whether newer titles are becoming shorter, longer, or staying consistent.

Research Question

Is there a relationship between a Netflix title's release year and its duration? For movies, the question is whether newer movies tend to be shorter or longer in minutes. For TV shows, the question is whether newer shows tend to have more or fewer seasons.

Hypothesis

Null hypothesis: There is no meaningful relationship between release year and title duration.
Alternative hypothesis: Release year and duration are related, with newer titles showing a different duration pattern than older titles.

Population of Interest

The population of interest is the full set of Netflix titles represented in the catalog, including both movies and TV shows. In practical terms, this means all titles Netflix has listed in the dataset, not just a small subgroup.

Sampling Method

This analysis uses the entire available Netflix catalog provided in the dataset, which is a census of the uploaded file rather than a sampled subset. That is the best available approach because it avoids sampling error and lets the report reflect every title in the file.

Bias Identification

A possible bias in this study is catalog bias: the dataset only includes titles that were available in the Netflix listing, so it does not represent every show or movie ever made. To minimize distortion, the analysis uses the full dataset as provided, keeps the movie and TV show analyses separate, and avoids mixing minutes and seasons into one misleading measure.

Survey Questions/Collected Data/Dataset

Although this project is not a survey, the dataset functions like structured collected data about Netflix titles. The key variables used are `show_id`, `type`, `title`, `release_year`, `duration`, `date_added`, `rating`, `listed_in`, and `country`. Release year gives the year each title originally came out, duration is recorded as minutes for movies and seasons for TV shows, type separates movies from TV shows, and `listed_in` shows the genre categories. The analysis focuses only on variables needed to study the relationship between release year and duration.

Number of samples used: 8,807 total titles, including 6,128 movies and 2,676 TV shows. For the hypothesis analysis, 6,128 movie records and 2,676 TV show records were usable after cleaning the duration and release year fields.

Analysis

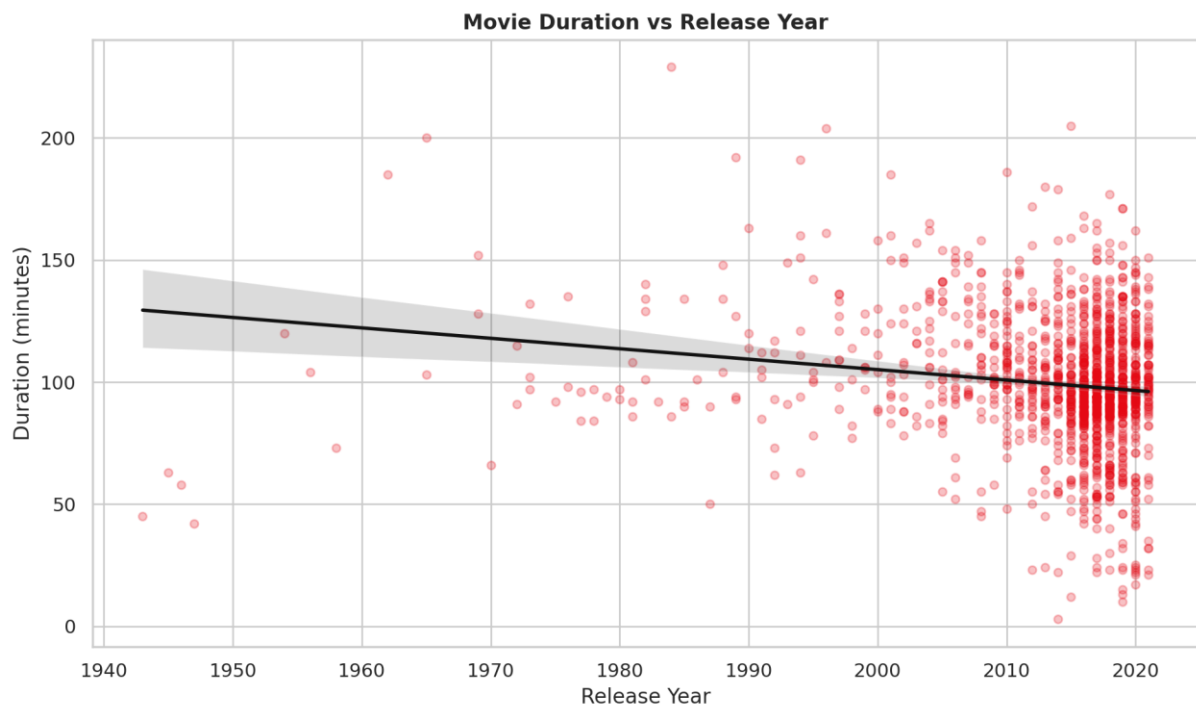
To analyze the data, descriptive statistics were calculated and the relationship between release year and duration was visualized with charts. The movie analysis uses duration in minutes, while the TV show analysis uses number of seasons. This separation is important because the two content types measure length differently.

Movie duration average: 99.6 minutes | median: 98.0 | mode: 90

TV show seasons average: 1.76 seasons | median: 1 | mode: 1

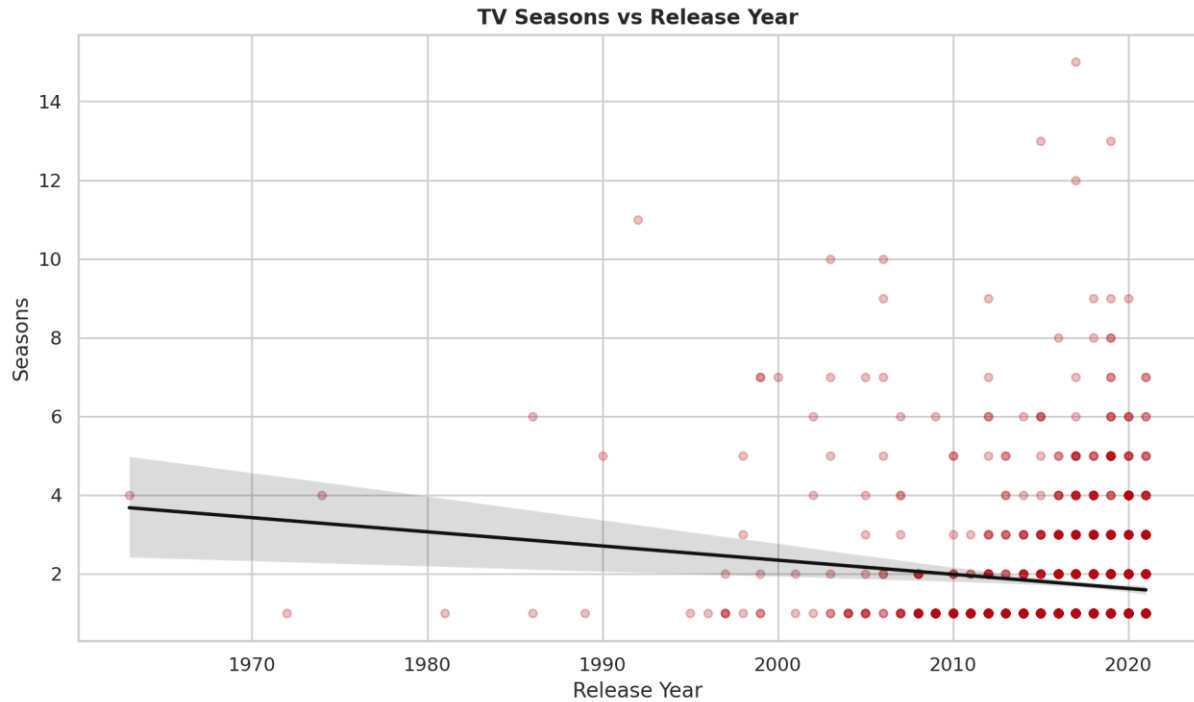
Correlation for movies: $r = -0.206$, $p\text{-value} < 0.001$. Correlation for TV shows: $r = -0.090$, $p\text{-value} < 0.001$.

Movie Duration vs Release Year



This chart shows how movie length changes as release year increases. The downward trend suggests that newer movies in the Netflix catalog tend to be slightly shorter on average than older ones. The relationship is statistically significant, but the slope is modest, so the change is not dramatic in everyday terms.

TV Seasons vs Release Year



This chart shows the relationship between release year and the number of seasons for TV shows. The pattern is much weaker than the movie chart, which means release year does not strongly predict how many seasons a TV show has. Even though the test is statistically significant, the practical relationship is small.

Hypothesis Testing Steps

Step 1: State the hypotheses. Null: release year and duration are not related. Alternative: release year and duration are related.

Step 2: Choose a significance level of 0.05.

Step 3: Run correlation and linear trend tests separately for movies and TV shows after cleaning the duration and release year fields.

Step 4: Compare the p-values to the significance level and interpret the direction of the relationship.

Conclusion

The results suggest that newer movies in the Netflix catalog tend to be slightly shorter than older movies. For TV shows, the relationship between release year and seasons is weak and should not be overstated. Overall, the hypothesis that release year and duration are related is supported more clearly for movies than for TV shows, but the effects are not large enough to claim a strong practical pattern across the entire catalog.

Any potential issues

One limitation is that movie duration and TV seasons are not the same kind of measurement, so they must be analyzed separately. Another issue is that the dataset is a catalog snapshot, not a random sample of all entertainment titles, so the findings describe Netflix's listed library rather than the whole industry. Some records also have missing or inconsistent date and duration values, which can reduce precision even after cleaning.